

Lakshya A Agrawal

Ph.D. Student, BAIR, Sky Lab, and Berkeley NLP
University of California, Berkeley

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🏠 Website 🐙 GitHub 🔗 LinkedIn 🎓 Google Scholar

EDUCATION

University of California, Berkeley

2024 - Present

Ph.D. in Computer Science

Advisors: Prof. Matei Zaharia and Prof. Dan Klein

IIIT-Delhi

2018 - 2022

B.Tech in Computer Science and Applied Mathematics (Department Rank: 1)

GPA: 9.55/10

WORK EXPERIENCE

Microsoft Research

AI4Code Research Fellow (Prev. Research Intern)

Jan 2022 - Aug 2024

Advisors: Dr. Aditya Kanade, Dr. Navin Goyal, Dr. Shuvendu Lahiri, Dr. Sriram Rajamani

- Proposed Monitor-Guided Decoding for repository-grounded code generation with certifiable correctness properties (valid method call sequences, typestates, arguments, absence of hallucinated symbols). **Accepted at NeurIPS '23.**
- 20-25% improvements in compilability of LLM generated code without any modification to the models.**

Very Large Scale Computing Laboratory, EPFL

Summer@EPFL Research Fellow

July 2021 - Dec 2021

Advisors: Dr. Endri Bezati, Prof. James Larus

- Developed `streamblocks-graalvm`, a CPU based runtime for `CAL` dataflow language, based on `Truffle/GraalVM`.
- Implemented IDE and debugger support (with code stepping) for the `CAL` programming language over `LSP` and `DAP`.

Google Summer of Code

Open Source Developer @ INCF

May 2019 - Aug 2019

Advisors: Dr. Dimitar Prodanov, Dr. Robert Dodier

- Developed `Pytranslate`, a transpiler in Common Lisp, that translates Maxima symbolic computation code to Python, including support for translating 2D and 3D plot functions. **Included as a part of all Maxima installations** since 2019.

PUBLICATIONS

📄 GEPA: Reflective Prompt Evolution Can Outperform Reinforcement Learning

Lakshya A Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziems, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alexandros G. Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, Omar Khattab

International Conference on Learning Representations (ICLR), 2026 **Oral**

10M+ downloads. Featured on *VentureBeat*. Core optimizer in *DSPy*, *MLflow*, *Pydantic AI*, *Comet ML*.

📄 optimize_anything: A Universal API for Optimizing any Text Parameter

Lakshya A Agrawal*, Donghyun Lee*, Wenjie Ma, Karim Elmaaroufi, Shangyin Tan, Sanjit A. Seshia, Koushik Sen, Dan Klein, Ion Stoica, Joseph E. Gonzalez, Omar Khattab, Alexandros G. Dimakis, Matei Zaharia

Technical Report, February 2026.

📄 Automatically Learning Skills for Coding Agents

Shangyin Tan*, Lakshya A Agrawal*, Rohit Sandadi, Dan Klein, Koushik Sen, Alexandros G. Dimakis, Matei Zaharia

Technical Report, February 2026.

📄 Multi-module GRPO: Composing Policy Gradients and Prompt Optimization for Language Model Programs

Noah Ziems*, Dilara Soylu*, Lakshya A Agrawal*, Isaac Miller, Liheng Lai, Chen Qian, Kaiqiang Song, Meng Jiang, Dan Klein, Matei Zaharia, Karel D'Oosterlinck, Christopher Potts, Omar Khattab

Preprint arXiv.

📄 Why Do Multi-Agent LLM Systems Fail?

Mert Cemri*, Melissa Z. Pan*, Shuyi Yang*, Lakshya A Agrawal*, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, Matei Zaharia, Joseph E. Gonzalez, Ion Stoica

Neural Information Processing Systems (NeurIPS), 2025 (Datasets & Benchmarks) **Spotlight**

📄 LangProBe: a Language Programs Benchmark

Shangyin Tan, Lakshya A Agrawal, Arnav Singhvi, Liheng Lai, Michael J Ryan, Dan Klein, Omar Khattab, Koushik Sen, Matei Zaharia

Findings of Empirical Methods in Natural Language Processing (EMNLP), 2025.

📄 Let the Barbarians In: How AI Can Accelerate Systems Performance Research

Audrey Cheng, Shu Liu, Melissa Pan, Zhifei Li, Shubham Agarwal, Mert Cemri, Bowen Wang, Alexander Krentsel, Tian Xia, Jongseok Park, Shuo Yang, Jeff Chen, **Lakshya A Agrawal**, Ashwin Naren, Shulu Li, Ruiying Ma, Aditya Desai, Jiarong Xing, Koushik Sen, Matei Zaharia, Ion Stoica
Preprint arXiv:2512.14806, 2025.

📄 Monitor-Guided Decoding of Code LMs with Static Analysis of Repository Context.

Lakshya A Agrawal, Aditya Kanade, Navin Goyal, Shuvendu Lahiri, Sriram Rajamani.
Neural Information Processing Systems (NeurIPS), 2023

📄 A SPARQL to Cypher Transpiler: Proposal and Initial Results. Extended Abstract

Lakshya A Agrawal, Nikunj Singhal, Raghava Mutharaju.
International Conference on Data Science and Management of Data (CODS-COMAD), 2022

AWARDS AND HONORS

- **Amazon AI PhD Fellow** 2025
- **Laude Slingshot** recipient (for GEPA) and **Open Research Resident** at Laude Institute 2025
- **First Position at Microsoft Global Hackathon** challenge on AI powered tools for developer productivity 2023
- (News Coverage) All Round Performance Medal for **Best Academic & cocurriculars** - CSAM batch 2018-22. 2022
- Summer@EPFL 2021 & 2020 scholarship (top 1% of applicants) 2020, 2021

SELECTED RESEARCH PROJECTS

GEPA: Reflective Prompt Evolution for Sample-efficient AI System Optimization Jan 2025 - Present

- Developed SOTA prompt optimizer that uses reflection to optimize LLM-based systems with high sample efficiency.
- Demonstrated that prompt learning can outperform GRPO by up to 20% while using up to 35× fewer rollouts.
- **Deployed in production by Databricks, OpenAI, Shopify, Dropbox, Uber among 50+ organizations.**

dspy.GRPO: First OSS multi-module, multi-step RL pipeline for LLM-agents March 2025 - Present

- Developed 1st GRPO pipeline for tuning modular agents, including complex compound AI Systems that compose multiple structured and specialized LM calls and tool invocations.
- Created server-client abstraction, decomposing GRPO policy gradient updates and multi-stage rollouts for efficiency.

ANNS-based KV retrieval for long-range attention over large code repos Aug 2022 - Oct 2022

- Augmented SantaCoder and CodeGen models to retrieve Key/Value values during attention from large vector stores using FAISS-based ANNS. Promising initial results of better code style match and lower symbol hallucinations.

SELECTED DEVELOPMENT PROJECTS

multilspy : Batteries included LSP client library in Python (HackerNews Frontpage) Jan 2024 - Present

- LSP client providing unified access to static analyses (type-directed completion, symbol definition, references) across 10 languages (Java, Python, Rust, C#, TypeScript, Go, etc.). **540+ stars, 100+ forks.**
- Open-sourced as part of Monitors4Codegen (280+ stars), the artifact for our NeurIPS '23 Monitor-Guided Decoding paper.
- **Deployed in production by several AI4Code startups.** multilspy makes it very easy to develop IDE tools for LLMs.

Spoon : Java Metaprogramming Library by INRIA Aug 2020 - Oct 2020

Advisor: Prof. Martin Monperrus

- OSS Contributions to Spoon (library for analyzing and transforming Java source code) to add new features and fix bugs.
- Implemented TextBlock support (JEP 355) through Eclipse JDT, thus completing Java 15 support in Spoon.

CovidReliefBot : Chatbot for COVID resource information Apr 2021 - May 2021

- Developed a chatbot to aid volunteers in resolving COVID related resource requests piloted with **up to 10,000 members.**
- Transcribed information images and aggregate relevant results from various data sources like Twitter, Google Sheets, etc.

INVITED TALKS

- **AI-Driven Code Performance Optimization for New Accelerators**
– AMD What's New In Labs Talk Series Jan 2025
- **Guiding Language Models of Code with Global Context using Monitors**
– Microsoft Research RiSE Group, Microsoft Research India Lab, Microsoft DevDiv July, Aug 2023
- **CAL Implementation in GraalVM**
– Very Large Scale Computing Lab, Data Center Systems Lab @ EPFL Sept 2021

ACADEMIC SERVICE

- **Conference Reviewer** - ICLR 2025, ISEC 2024, NeurIPS 2025 (Recognized as **Top Reviewer**)